Specifications for the Description of Wheat Germplasm Resources

Author-prepared English reference translation

This reference translation follows the Chinese crop descriptor specification used for the CropMaestro study. It is not an official English edition issued by the source platform. Original section numbers and category-code mappings are retained.

1 Scope

This specification specifies the descriptors for wheat germplasm resources and their classification criteria.

This specification applies to the collection, documentation and conservation of wheat germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative references

The provisions of the following documents, through reference in this specification, constitute provisions of this specification. For dated references, subsequent amendments (excluding corrections) or revisions do not apply to this specification. However, parties to agreements based on this specification are encouraged to investigate whether the latest editions of these documents can be used. For undated references, the latest edition applies to this specification.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659 Codes for the Names of Countries and Regions of the World

GB/T 2260 Codes for the Administrative Divisions of the People's Republic of China

GB/T 12404 Codes for the Administrative Affiliation of Organizations

GB/T 3543-1995 Rules for Agricultural Seed Testing

GB/T19557.2-2004 Guidelines for DUS Testing of New Wheat Varieties

3 Terms and definitions

3.1 Wheat

Annual or winter-annual herbaceous plants of the genus Triticum L. in the family Gramineae. Approximately 27 species occur worldwide; common wheat, Triticum aestivum L., is the species most widely used in agricultural production.

3.2 Wheat germplasm resources

Plant material capable of contributing germplasm to wheat. Wheat germplasm resources include species of Triticum and related genera, both wild and cultivated. They include landraces, developed varieties and introduced varieties, as well as lines with particular desirable traits, mutants, male-sterile materials and aneuploids.

3.3 Basic information

Descriptive information on the basic attributes of wheat germplasm resources, including the national identification number, germplasm name, place of origin, holding institution, breeding institution, year of development and breeding method.

3.4 Morphological and biological characteristics

The morphological and biological characteristics of wheat germplasm resources, together with economic traits and traits associated with crop management. In practice, these comprise morphological traits (such as awn form, glume and lemma colour, spike shape, grain colour and grain shape), developmental traits (such as jointing date, heading date and maturity date), yield traits (such as grains per spike and thousand-grain weight), and traits associated with crop management (such as sowing date and emergence date).

3.5 Quality characteristics

The nutritional quality of wheat grains, including crude protein content, lysine content, wet gluten content and sedimentation value.

3.6 Abiotic stress resistance

The ability of wheat germplasm resources to adapt to or withstand abiotic stresses, including drought resistance, salinity tolerance, cold resistance and waterlogging tolerance.

3.7 Disease resistance

The ability of wheat germplasm resources to resist pathogen invasion and spread. Immunity denotes resistance to invasion; a high degree of disease resistance denotes restriction of pathogen spread; and weak resistance denotes susceptibility. The principal diseases affecting wheat germplasm resources include stripe rust, leaf rust, stem rust, powdery mildew, Fusarium head blight, root rot, sharp eyespot, yellow dwarf disease and take-all.

3.8 Insect resistance

The ability of wheat germplasm resources to resist damage by insect pests. The principal insect pests affecting wheat germplasm resources include aphids and wheat midges.

3.9 Wheat growth cycle

On the basis of the characteristics of its development, the wheat life cycle can be divided into 12 growth stages: emergence, three-leaf stage, tillering, overwintering, spring green-up, pseudostem erection, jointing, booting, heading, flowering, grain filling and maturity. Winter wheat generally requires 230-270 d, whereas spring wheat requires 90-150 d.

On the basis of organ formation, several consecutive growth stages can be combined into a growth phase. The seedling phase extends from emergence to the pseudostem-erection stage and consists mainly of vegetative growth, particularly root growth, leaf growth and tillering. The intermediate phase extends from pseudostem erection to flowering and involves concurrent vegetative and reproductive growth: roots, stems and leaves grow while the spike differentiates and develops. The late phase extends from flowering to maturity, is also called the grain-formation phase, and consists mainly of reproductive growth.

4 Basic information

4.1 National identification number

The unique identifier of the germplasm. For common wheat germplasm originating in China, the number consists of "ZM" followed by a 6-digit sequential number. For common wheat germplasm originating outside China, it consists of "MY" followed by a 6-digit sequential number.

4.2 Genebank accession number

The number assigned to wheat germplasm in the National Long-Term Crop Germplasm Genebank. It consists of "I1B" followed by a 5-digit sequential number.

4.3 Introduction number

The number assigned to wheat germplasm when it is introduced from abroad.

4.4 Collecting number

The number assigned to wheat germplasm during field collection.

4.5 Germplasm name

The Chinese name of the wheat germplasm.

4.6 Foreign-language germplasm name

The foreign-language name of wheat germplasm introduced from abroad, or the Chinese Pinyin name used for the international exchange of domestic germplasm.

4.7 Family

Gramineae.

4.8 Genus

Triticum L.

4.9 Scientific name

The scientific name of common wheat is Triticum aestivum L.

4.10 Country of origin

The name of the country, region or international organization of origin of the wheat germplasm resource.

4.11 Province of origin

For domestic wheat germplasm, the name of the province of origin; for germplasm introduced from abroad, the name of the first-level administrative division in the country of origin.

4.12 Locality of origin

The names of the county, township and village of origin of domestic wheat germplasm.

4.13 Elevation

The elevation of the place of origin of the wheat germplasm. Unit: m.

4.14 Longitude

The longitude of the place of origin of the wheat germplasm, expressed in degrees (°) and minutes (′). The format is DDDFF, where DDD denotes degrees and FF denotes minutes.

4.15 Latitude

The latitude of the place of origin of the wheat germplasm, expressed in degrees (°) and minutes (′). The format is DDFF, where DD denotes degrees and FF denotes minutes.

4.16 Source locality

For wheat germplasm introduced from abroad, the name of the supplying country, region or international organization; for domestic germplasm, the name of the supplying province or county.

4.17 Holding institution

The name of the institution that held the wheat germplasm before its submission to the National Long-Term Crop Germplasm Genebank.

4.18 Accession number at the holding institution

The germplasm number assigned by the institution that originally held the wheat germplasm.

4.19 Pedigree

The genetic relationships of bred wheat varieties or lines.

4.20 Breeding institution or breeder

The name of the institution or individual that developed the wheat variety or line.

4.21 Year of development

The year in which the wheat variety or line was successfully developed.

4.22 Breeding method

The breeding method used to develop the wheat variety or line.

4.23 Germplasm type

Wheat germplasm is classified into 6 types.

1	Wild germplasm
2	Landrace
3	Bred variety
4	Breeding line
5	Genetic stock
6	Other

4.24 Image

The image filename for the wheat germplasm. The image format is .jpg.

4.25 Observation site

The name of the site where the morphological and biological characteristics of the wheat germplasm are observed.

5 Morphological and biological characteristics

5.1 Winter and spring wheat

By sowing season, wheat is classified into winter wheat and spring wheat. Winter wheat refers to varieties sown in autumn and harvested in the summer of the following year. Spring wheat refers to varieties sown in spring and harvested in summer or autumn of the same year.

1	Winter wheat
2	Spring wheat

5.2 Winter-spring growth habit

The degree and duration of low temperature required for wheat germplasm to complete vernalization during the seedling stage.

1	Winter
2	Weak winter
3	Spring
4	Facultative

5.3 Sowing date

The date of field sowing, recorded as year, month and day in the format YYYYMMDD.

5.4 Emergence date

The date after sowing on which at least 50% of the seedlings in the plot have emerged 2-3 cm above the soil surface, recorded as year, month and day in the format YYYYMMDD.

5.5 Spring green-up date

For autumn-sown winter-type germplasm in northern regions, the date in the following spring when overwintered leaves regain their green colour and new leaves emerge, after the aboveground leaves have withered or lost their green colour because of winter cold. Record the date as year, month and day in the format YYYYMMDD.

5.6 Jointing date

The date on which the stems of 50% of the plants in the plot have elongated to 3-4 cm and the first internode extends 1.5-2.0 cm above the soil surface. Record the date as year, month and day in the format YYYYMMDD.

5.7 Heading date

The date on which the spikes of 50% of the plants in the plot have emerged from the flag-leaf sheaths. Record the date as year, month and day in the format YYYYMMDD.

5.8 Flowering date

The date on which the spikes of 50% of the plants in the plot are flowering or have exposed anthers. Record the date as year, month and day in the format YYYYMMDD.

5.9 Maturity date

The date on which the grains reach the dough stage (wax ripeness), recorded as year, month and day in the format YYYYMMDD.

5.10 Maturity class

Earliness or lateness of maturity relative to locally grown mid-maturing varieties cultivated over a large area.

1	Very early
2	Early
3	Medium
4	Late
5	Very late

5.11 Total growth duration

The number of days from sowing to maturity. Unit: d.

5.12 Photoperiod response

The response of wheat germplasm plants to day length.

1	Insensitive
2	Intermediate
3	Sensitive

5.13 Seed dormancy period

After maturity, seeds require a period of after-ripening before they can germinate; this interval is termed the dormancy period. The length of seed dormancy differs considerably among varieties.

1	Short
2	Medium
3	Long

5.14 Coleoptile colour

The colour of the young coleoptile when it has emerged approximately 2 cm above the soil surface.

1	Green
2	Purple

5.15 Seedling growth habit

The attitude of the leaves of wheat germplasm at peak tillering (see Figure 1).

1	Erect
2	Semi-prostrate
3	Prostrate

Figure 1. Seedling growth habit



1: Erect; 2: Semi-prostrate; 3: Prostrate.

5.16 Seedling colour

The colour of the seedlings.

1	Light green
2	Green
3	Dark green

5.17 Seedling leaf length

The length of the seedling leaf blade. Unit: cm.

5.18 Seedling leaf width

The width of the seedling leaf blade. Unit: cm.

5.19 Leaf pubescence

The presence or absence of hairs on the leaf surface.

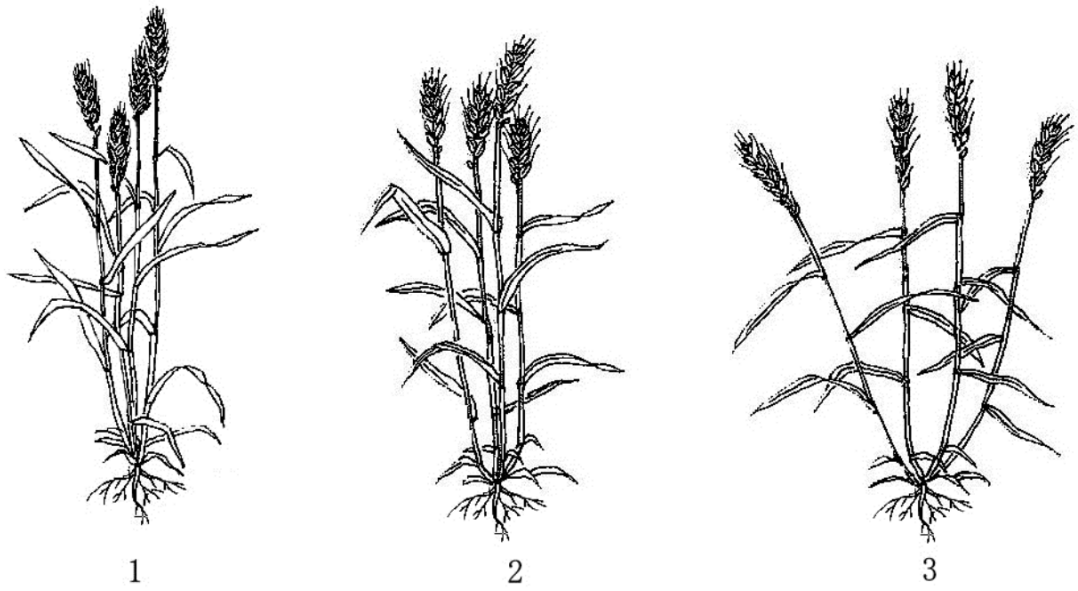
0	Absent
1	Present

5.20 Plant architecture

The degree of compactness or spreading of the main stem and tiller stems after heading (see Figure 2).

1	Compact
2	Intermediate
3	Spreading

Figure 2. Plant architecture



1: Compact; 2: Intermediate; 3: Spreading.

5.21 Leaf attitude

Leaf shape, the angle between the leaf and stem, and the extent of leaf spreading or drooping (see Figure 3).

1	Erect
2	Horizontal
3	Drooping

Figure 3. Leaf attitude



1: Erect; 2: Horizontal; 3: Drooping.

5.22 Flag-leaf length

The flag leaf is the first leaf immediately below the spike. Flag-leaf length is measured from the base of the leaf blade to its tip.

1	Short
2	Medium
3	Long

5.23 Flag-leaf width

The flag leaf is the first leaf immediately below the spike. Flag-leaf width is the width at the middle of the leaf blade.

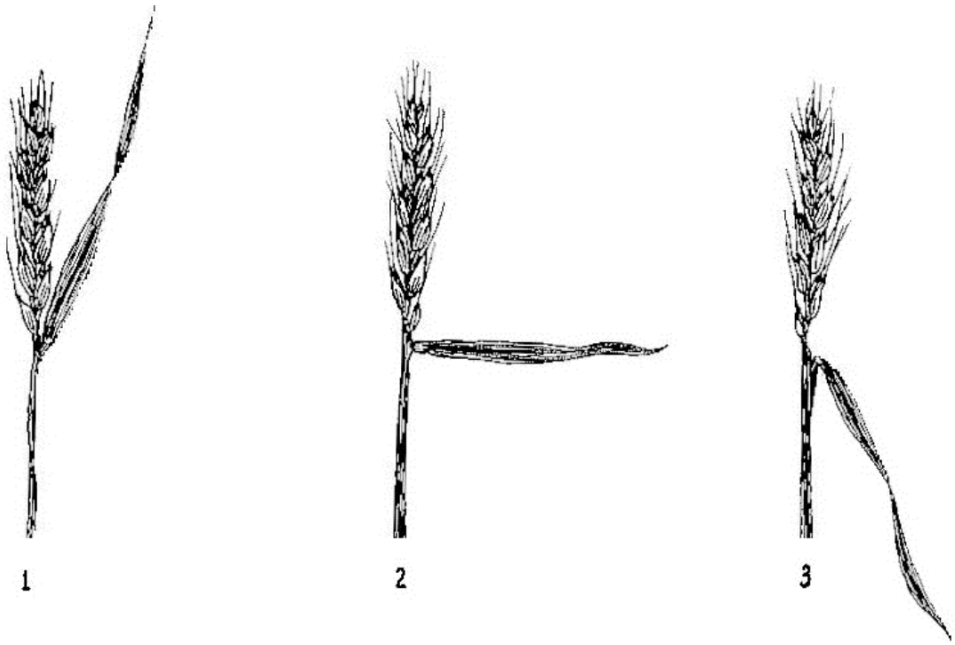
1	Narrow
2	Medium
3	Broad

5.24 Flag-leaf angle

The angle between the flag leaf and the stem below the spike (see Figure 4).

1	Erect
2	Horizontal
3	Drooping

Figure 4. Flag-leaf angle



1: Erect; 2: Horizontal; 3: Drooping.

5.25 Auricle colour

The colour of the auricles, the projections extending from either side of the leaf base.

1	Green
2	Purple

5.26 Anther colour

The anther is an important component of the stamen. Two anther colours are distinguished.

1	Yellow
2	Purple

5.27 Spike glaucousness

The presence and intensity of the powdery wax coating on the spike surface from flowering to grain filling.

0	Absent
1	Light
2	Heavy

5.28 Stem glaucousness

The presence and intensity of the powdery wax coating on the stem surface from flowering to grain filling.

0	Absent
1	Light
2	Heavy

5.29 Leaf glaucousness

The presence and intensity of the powdery wax coating on the leaf surface from flowering to grain filling.

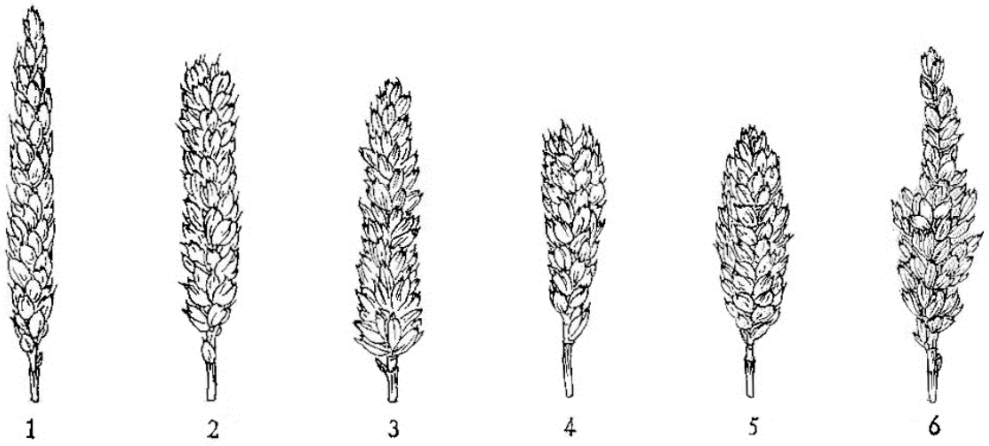
0	Absent
1	Light
2	Heavy

5.30 Spike shape

The shape of the spike at maturity (see Figure 5).

1	Fusiform
2	Rectangular
3	Conical
4	Clavate
5	Elliptical
6	Branched

Figure 5. Spike shape



1: Fusiform; 2: Rectangular; 3: Conical; 4: Clavate; 5: Elliptical; 6: Branched.

5.31 Stem colour

The colour of the stem at the dough stage (wax ripeness).

1	Yellow
2	Purple

5.32 Awn form

The presence or absence, length and degree of curvature of the awns, which are extensions of the lemma apex (see Figure 6).

0	Absent
1	Short
2	Long
3	Hooked
4	Short and curved
5	Long and curved

Figure 6. Awn form

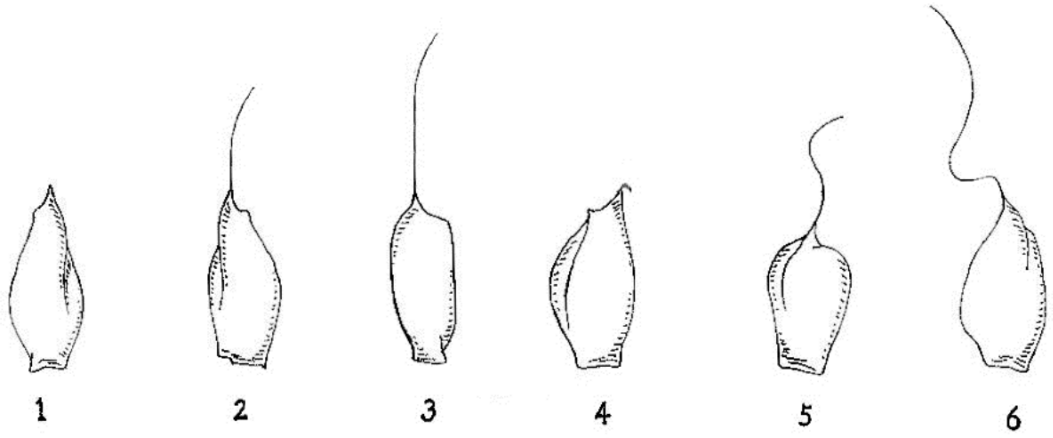


Illustration 1: absent (code 0); 2: short (code 1); 3: long (code 2); 4: hooked (code 3); 5: short and curved (code 4); 6: long and curved (code 5). The illustration numbers 1-6 correspond respectively to descriptor codes 0-5.

5.33 Awn colour

At maturity, awn colour usually matches glume and lemma colour. Awn colour is generally not recorded; only distinctive colours, such as black awns, are noted.

5.34 Glume and lemma colour

The colour of the glumes and lemmas at maturity.

1	White
2	Red
3	Black
4	White with black markings or margins
5	Red with black markings or margins

5.35 Glume pubescence

Whether the glumes are pubescent. Record only cases in which the glumes bear hairs.

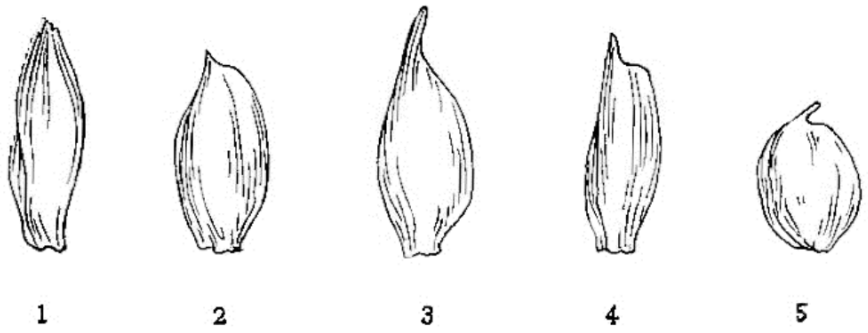
0	Absent
1	Present

5.36 Glume shape

The shape of the glume in lateral view, assessed on glumes from the middle of the main spike (see Figure 7).

1	Oblong
2	Elliptical
3	Ovate
4	Rectangular
5	Round

Figure 7. Glume shape



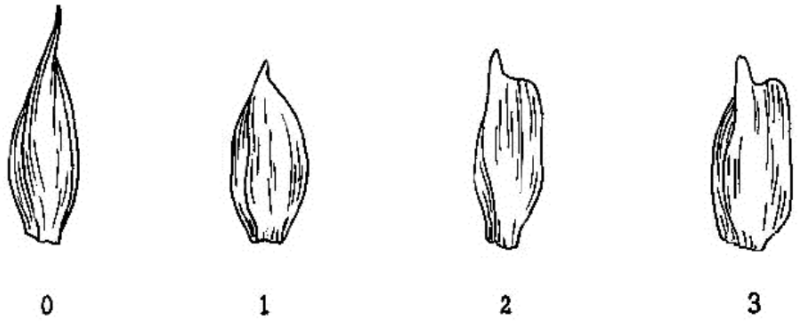
1: Oblong; 2: Elliptical; 3: Ovate; 4: Rectangular; 5: Round.

5.37 Glume shoulder shape

The shape of the glume shoulder, assessed on glumes from the middle of the main spike (see Figure 8).

0	Absent
1	Sloping
2	Square
3	Elevated

Figure 8. Glume shoulder shape



0: Absent; 1: Sloping; 2: Square; 3: Elevated.

5.38 Glume beak shape

The shape of the glume tip, assessed on glumes from the middle of the main spike (see Figure 9).

1	Obtuse
2	Acute
3	Beak-shaped

Figure 9. Glume beak shape



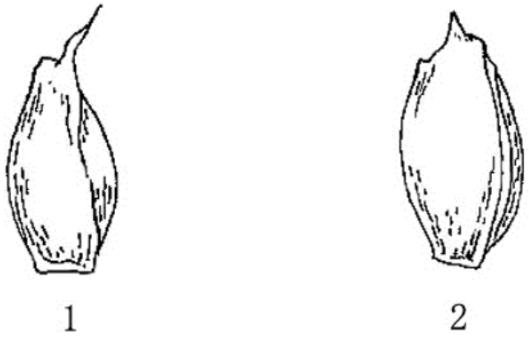
1: Obtuse; 2: Acute; 3: Beak-shaped.

5.39 Glume keel prominence

The raised, keel-like ridge in the middle of the glume is termed the glume keel. In some varieties, the keel is serrated (see Figure 10).

1	Indistinct
2	Distinct

Figure 10. Glume keel prominence



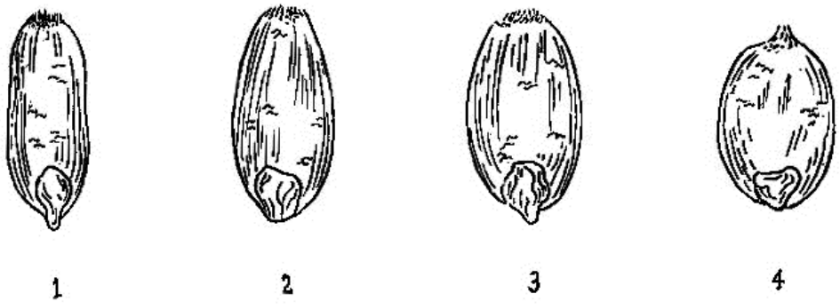
1: Indistinct; 2: Distinct.

5.40 Grain shape

The shape of fully mature grains (see Figure 11).

1	Oblong
2	Ovate
3	Elliptical
4	Round

Figure 11. Grain shape



1: Oblong; 2: Ovate; 3: Elliptical; 4: Round.

5.41 Ventral crease

The depth and width of the groove on the ventral surface of the grain.

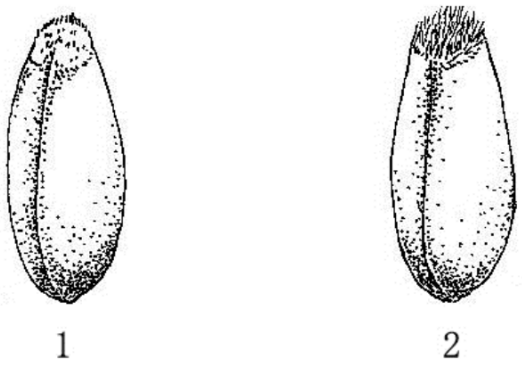
1	Shallow
2	Deep

5.42 Grain brush hair abundance

The abundance of hairs at the grain apex (see Figure 12).

1	Sparse
2	Abundant

Figure 12. Grain brush hair abundance



1: Sparse; 2: Abundant.

5.43 Grain colour

The colour of mature grains.

1	White
2	Red
3	Black-purple
4	Blue or green

5.44 Grain texture

The hardness or softness of the endosperm in fully mature grains.

1	Soft
2	Semi-hard
3	Hard

5.45 Grain size

The size of fully mature grains, classified according to thousand-grain weight.

1	Small
2	Medium
3	Large
4	Very large

5.46 Grain plumpness

The degree of filling of fully mature grains.

1	Poorly filled
2	Intermediate
3	Plump

5.47 Grain uniformity

The degree of size variation among grains, assessed visually for uniformity.

1	Nonuniform
2	Intermediate
3	Uniform

5.48 Plant height

The height of the plant, measured around the milk-ripe stage from the soil surface to the tip of the spike, excluding the awns. Unit: cm.

5.49 Plant uniformity

The uniformity of plant height and of the heights of the main and tiller spikes from heading to maturity.

1	Nonuniform
2	Intermediate
3	Uniform

5.50 Tiller number

The total number of tillers per plant, including productive and nonproductive tillers. Unit: tillers.

5.51 Productive tiller number

A productive tiller is one that produces a spike and sets grain. Count the productive tillers per plant. Unit: tillers.

5.52 Spike length

The length of the spike from its base to its tip, excluding the awns. Unit: cm.

5.53 Spikelet density

The average number of spikelets borne on 10 cm of the rachis.

1	Sparse
2	Intermediate
3	Dense
4	Very dense

5.54 Spikelets per spike

The total number of spikelets borne on one spike, including sterile spikelets. Unit: spikelets.

5.55 Sterile spikelets per spike

The number of spikelets at the lower and upper portions of a spike that do not set grain. Unit: spikelets.

5.56 Grains per spikelet

The number of grains set in the spikelet bearing the most grains in the middle of the spike. Unit: grains.

5.57 Grains per spike

The total number of grains set per spike. Unit: grains.

5.58 Grain weight per spike

The total weight of grains per spike. Unit: g.

5.59 Thousand-grain weight

The weight of 1000 dried grains. Unit: g.

5.60 Biological yield per plant

The weight of the roots, stems, leaves and spikes of a plant. Unit: g.

5.61 Shattering

The degree of grain shedding under strong winds after the grains are fully mature.

1	Readily shattering
2	Intermediate
3	Firmly retained

5.62 Lodging resistance

The degree of plant recovery after lodging caused by strong wind and rain during the period from heading to maturity.

1	Strong
2	Medium
3	Weak
4	Very weak

6 Quality characteristics

6.1 Seed moisture content

The mass of water in the seeds as a percentage of total seed mass. Expressed as %.

6.2 Test weight

The mass of wheat grains per unit volume. Unit: g/L.

6.3 Grain hardness

The hardness or softness of wheat grains, expressed by the grinding time method (GT). Unit: s.

6.4 Crude protein content

The mass of crude protein in the grains as a proportion of grain mass. Expressed as %.

6.5 Lysine content

The mass of lysine in the grains as a proportion of grain mass. Expressed as %.

6.6 SDS sedimentation value

The volume of flocculent sediment formed within a standard time interval by flour suspended in a sodium dodecyl sulfate-lactic acid solution of a specified concentration. Unit: mL.

6.7 Wet gluten content

The wet gluten content of wheat flour at a moisture content of 14%. Expressed as %.

7 Abiotic stress resistance

7.1 Drought resistance at germination

The ability of wheat germplasm to withstand water stress during seed germination.

1	Very strong (HR)
2	Strong (R)
3	Medium (MR)
4	Weak (S)
5	Very weak (HS)

7.2 Drought resistance at the seedling stage

The ability of wheat germplasm to withstand soil or atmospheric drought during the seedling stage.

1	Very strong (HR)
2	Strong (R)
3	Medium (MR)
4	Weak (S)
5	Very weak (HS)

7.3 Drought resistance throughout the growth period

The ability of wheat plants to withstand soil or atmospheric drought throughout the growth period.

1	Very strong (HR)
2	Strong (R)
3	Medium (MR)
4	Weak (S)
5	Very weak (HS)

7.4 Salinity tolerance at germination

The ability of wheat to tolerate stress caused by high salt concentrations in the soil during seed germination.

1	Highly tolerant (HR)
2	Tolerant (R)
3	Moderately tolerant (MR)
4	Sensitive (S)
5	Highly sensitive (HS)

7.5 Salinity tolerance at the seedling stage

The ability of wheat to tolerate stress caused by high salt concentrations in the soil during the seedling stage.

1	Highly tolerant (HR)
2	Tolerant (R)
3	Moderately tolerant (MR)
4	Sensitive (S)
5	Highly sensitive (HS)

7.6 Salinity tolerance throughout the growth period

The ability of wheat plants to tolerate stress caused by high salt concentrations in the soil throughout the growth period.

1	Highly tolerant (HR)
2	Tolerant (R)
3	Moderately tolerant (MR)
4	Sensitive (S)
5	Highly sensitive (HS)

7.7 Cold resistance

The ability of winter wheat seedlings in northern regions to withstand winter cold.

1	Very strong (HR)
3	Strong (R)
5	Medium (MR)
7	Weak (S)
9	Very weak (HS)

7.8 Waterlogging tolerance

The ability of wheat to tolerate stress caused by excessive soil moisture.

1	Strong
2	Medium
3	Weak

7.9 Resistance to sprouting in the spike

The ability, once maturity begins, to resist germination of grains in the spike during prolonged rainy weather, either in the field or after harvest.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8 Disease and insect resistance

8.1 Stripe rust resistance

The degree of resistance of wheat plants to infection by the stripe rust pathogen, Puccinia striiformis Westend.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.2 Leaf rust resistance

The degree of resistance of wheat plants to infection by the leaf rust pathogen, Puccinia triticina Eriks.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.3 Stem rust resistance

The degree of resistance of wheat plants to infection by the stem rust pathogen, Puccinia graminis Pers.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.4 Powdery mildew resistance

The degree of resistance of wheat plants to infection by the powdery mildew pathogen, Blumeria graminis (DC.) E.O. Speer.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.5 Fusarium head blight resistance

The degree of resistance of wheat plants to infection by Fusarium spp., the pathogens causing Fusarium head blight.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.6 Root rot resistance

The degree of resistance of wheat plants to infection by the root rot pathogen, Bipolaris sorokiniana (Sacc.) Shoemaker.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.7 Sharp eyespot resistance

The degree of resistance of wheat plants to infection by the sharp eyespot pathogen, Rhizoctonia cerealis Van der Hoeven.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.8 Yellow dwarf disease resistance

The degree of resistance of wheat plants to infection by Barley yellow dwarf virus.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.9 Take-all resistance

The degree of resistance of wheat plants to infection by the take-all pathogen, Gaeumannomyces graminis (Saccardo) von Arx & Olivier.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.10 Aphid resistance

The degree of resistance of wheat to damage by cereal aphids: the English grain aphid, Sitobion avenae (Fabricius); the greenbug, Schizaphis graminum (Rondani); the bird cherry-oat aphid, Rhopalosiphum padi (Linnaeus); and the rose-grain aphid, Metopolophium dirhodum (Walker).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.11 Wheat midge resistance

The degree of resistance of wheat plants to damage by the orange wheat blossom midge, Sitodiplosis mosellana (Gehin).

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

9 Other characteristics

9.1 Hybrid wheat

The first filial generation produced by crossing a restorer line with a male-sterile line and exhibiting heterosis.

1	Three-line hybrid wheat
2	Two-line hybrid wheat
3	Hybrid wheat produced using chemical hybridizing agents

9.2 Wheat aneuploids

Wheat germplasm resources whose chromosome number deviates from an exact multiple of the basic chromosome number. These resources include individual plants or complete sets of stocks in which individual chromosomes or chromosome arms occur in numbers greater or smaller than normal.

1	Primary aneuploids
2	Secondary aneuploids

9.3 Karyotype

The karyotype is the complete set of chromosome characteristics observed at metaphase of cell division, represented by a formula describing chromosome number, size, morphology and structural features.

9.4 Near-isogenic lines

Near-isogenic lines (NILs) are lines developed through a series of backcrosses that have identical or similar genetic backgrounds and differ only in particular chromosome regions.

In breeding practice, a donor parent carrying a gene for a marker trait is crossed with a recurrent parent, followed by repeated backcrossing. In each generation, only individuals carrying the target gene are selected for backcrossing to the recurrent parent. This produces lines with the same genetic background as the recurrent parent except for the target gene.

9.5 Recombinant inbred lines

Two varieties are crossed to produce F1 plants, which are self-pollinated to produce the F2 generation. Several hundred to a thousand or more individual plants are randomly selected from the F2 generation and self-pollinated. Only one seed per plant is sown in each subsequent generation, and this process continues to F6-F8, producing several hundred recombinant inbred lines.

A population developed from the F2 generation through repeated self-pollination and single-seed descent (SSD), resulting in relatively homozygous progeny genomes, is termed a recombinant inbred line (RIL) population.

9.6 Doubled-haploid population

A homozygous, permanent population produced by in vitro anther culture of F1 plants or by special techniques that induce haploid plants, followed by chromosome doubling. This is termed a doubled-haploid (DH) population and is commonly used in wheat QTL studies.

9.7 DNA fingerprints and molecular markers

The DNA fingerprint profiles of wheat germplasm, together with the types and characteristic parameters of molecular markers associated with important traits.

9.8 Remarks

Detailed explanations of special descriptors or special codes for wheat germplasm.

Translator's notes on the source

5.32: In Figure 6, illustration numbers 1-6 correspond respectively to descriptor codes 0-5.

5.35: The source instructs that only pubescent glumes are recorded, while also assigning code 0 to absence.

5.43: The source colour term 'qing' does not distinguish blue from green.

5.62: The source defines this descriptor by plant recovery after lodging.

8.3: The repeated author abbreviation 'Pers. Pers.' in the source has been reduced to 'Pers.'.

8.9: The source author citation '(Sacc.) Arx. olivcr' has been standardized to '(Saccardo) von Arx & Olivier', following EPPO entry GAEUGR; the species name has been retained.

8.11: The source author name 'Genin' has been standardized to '(Gehin)', following EPPO entry SITDMO.

9.5: The source's English expansions of SSD and RILs have been standardized as 'single-seed descent' and 'recombinant inbred lines'.